

VISITOR APPROVAL ENTRY SYSTEM

A full-stack visitor registration and approval platform — React, Node/Express & MySQL

Database Systems Engineering and Distributed Backend Development (25CS1302E)

Y25 · 2026–2027 · Trimester 4 | Koneru Lakshmaiah Education Foundation

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Abstract

- 1 What it is**

A digital gate-security app for apartment/office visitor entry, with separate Guard and Tenant portals.
- 2 Problem solved**

Replaces manual visitor registers and unverified walk-ins with a logged, approval-based entry process.
- 3 Core flow**

Guard registers a visitor → request sent to the tenant → tenant approves/denies → guard sees the result.
- 4 Tech used**

React (Vite) frontend, Node.js/Express REST API, MySQL database with bcrypt-hashed passwords.
- 5 Benefit**

Faster, traceable visitor handling with a permanent visit log and role-based access.

Problem Statement

Manual registers

Paper logs are slow, illegible, and easy to falsify or lose.

No approval step

Walk-in visitors often enter before the resident/host confirms them.

Poor traceability

No searchable history of who visited, when, or why.

No absence handling

Manual systems can't verify a visitor when the host is away.

A digital, role-based approval workflow is needed to close these gaps.

Objectives

- 1 Register visitor details digitally (name, phone, purpose, host block/flat)
- 2 Route each visit request to the tenant for approval or rejection
- 3 Support pre-approval of expected guests, so entry is instant
- 4 Allow tenants to schedule expected visits in advance
- 5 Maintain a permanent, searchable visit log for every entry
- 6 Provide secure, role-based login for guards and tenants

Proposed System

VAES is a two-portal web app (Guard / Tenant) backed by a REST API and MySQL database.

1

Authentication

Login & signup for tenants/guards; bcrypt password hashing

2

Security Module

Guard dashboard: register visitors, view sent requests

3

Tenant Module

Tenant dashboard: approvals, pre-approvals, schedules

4

Visitor Management

Visitor entry form; via Pre-Approved / Schedule / Direct

5

Approval Management

Tenant approves/denies pending visitor requests

6

Visit Logs

Append-only historical log of every visitor record

Future Scope

Not implemented in the current project — proposed enhancements only:

- 1 Facial Recognition – Verify visitors using facial identification
- 2 QR Code Verification – Provide quick and secure visitor entry.
- 3 Real-Time Notifications – Notify tenants instantly about visitor requests
- 4 Mobile application (Android/iOS)
- 5 Cloud deployment & scaling

Technology Stack

FRONTEND

React 18 + Vite

BACKEND / API

Node.js, Express.js

DATABASE

MySQL (mysql2 driver)

SECURITY

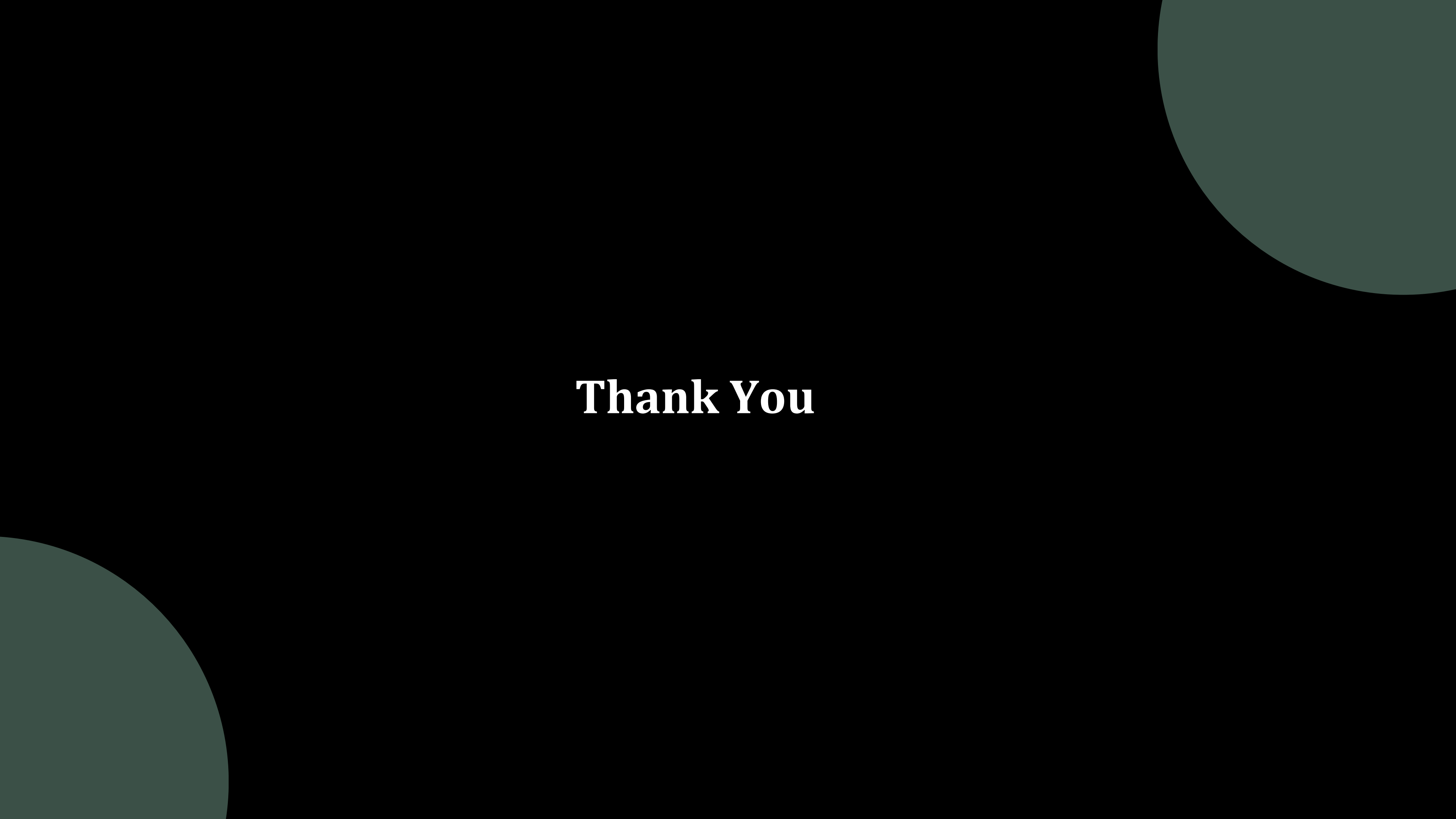
bcryptjs password hashing

CONFIG

dotenv, cors

DEV TOOLS

**VS Code, MySQL Workbench,
npm**



Thank You